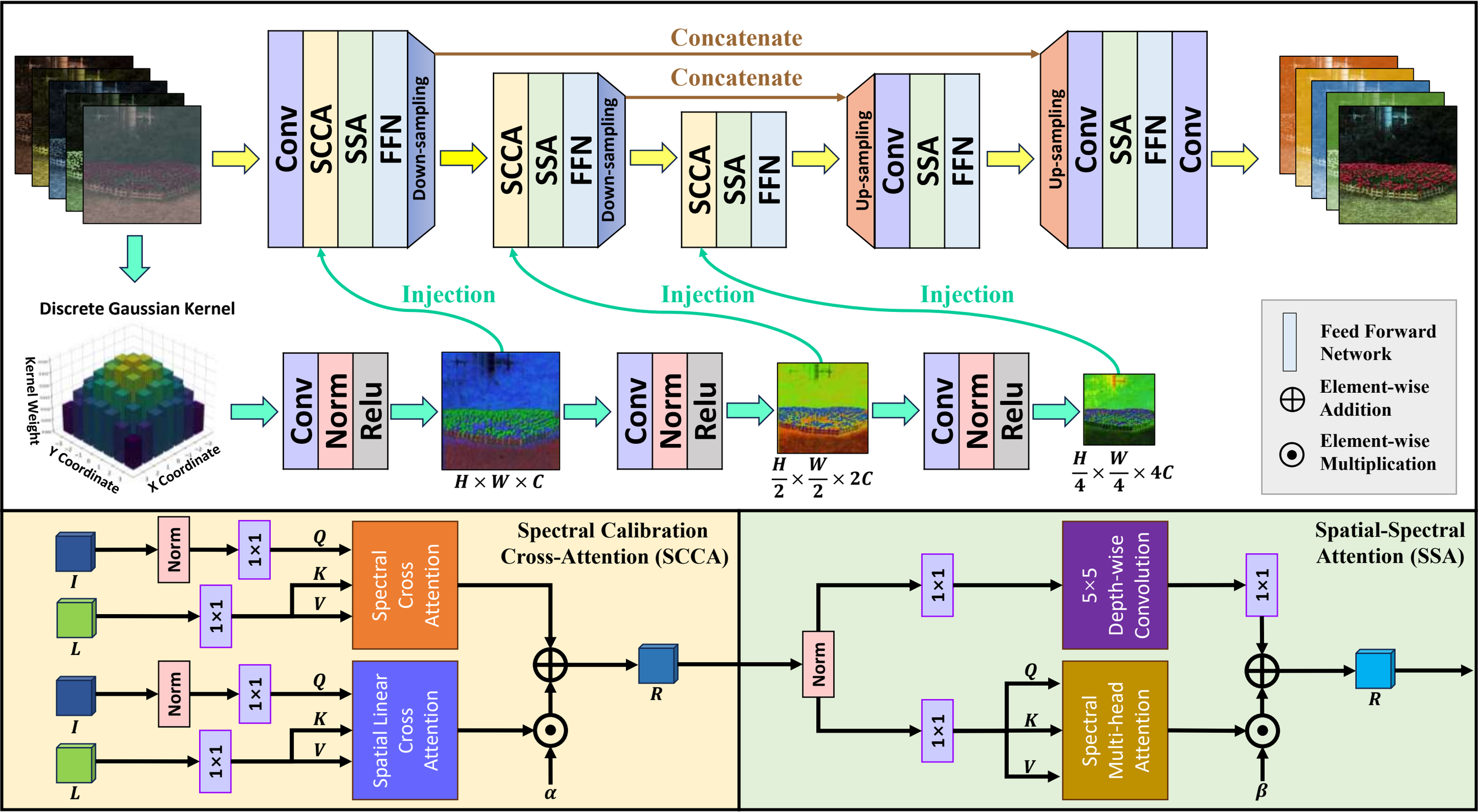

Concatenate
Concatenate
Conv
SCCA
SSA
FFN
Down-sampling
SCCA
SSA
FFN
Down-sampling
SCCA
SSA
FFN
Up-sampling
Conv
SSA
FFN
Up-sampling
Conv
SSA
FFN
Conv
Discrete Gaussian Kernel
Kernel Weight
Y Coordinate
X Coordinate
Injection
Injection
Injection
Conv
Norm
Relu
Conv
Norm
Relu
Conv
Norm
Relu
$H \times W \times C$
$\frac{H}{2} \times \frac{W}{2} \times 2C$
$\frac{H}{4} \times \frac{W}{4} \times 4C$
Feed Forward Network
Element-wise Addition
Element-wise Multiplication
Spectral Calibration Cross-Attention (SCCA)
Norm
1×1
Q
K
V
Spectral Cross Attention
Spatial Linear Cross Attention
I
L
R
α
Spatial-Spectral Attention (SSA)
Norm
1×1
5×5 Depth-wise Convolution
Spectral Multi-head Attention
β